CO2 Assessment Tool 1

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Representing Interacting Genes as a Network (4 Marks)

Concept: A network (graph) is a mathematical model G = (V, E). **Nodes (vertices)** represent biological entities and **edges (links)** represent the relationships between them. Network representation turns a long list of gene pairs into a structure that can be visualized and analyzed.

Representation:

Component	Biological meaning (breast cancer example)
Node	A gene or its protein product (e.g., BRCA1, BRCA2, TP53, ERBB2, ESR1, PIK3CA)
Edge	An interaction between two genes (physical binding, co-expression, regulation, or genetic interaction)
Node attributes	Expression fold change, mutation status, subtype, GO term
Edge attributes	Confidence score, interaction type, data source

Steps:

1. **Prepare the dataset** as an interaction table with columns: *Gene A, Gene B, interaction type, score*.
2. **Define each unique gene as a node.** Duplicate gene names are merged into one node.
3. **Define each interacting pair as an edge.** For example, the row "BRCA1 – TP53" becomes a link between those two nodes.
4. **Choose the edge type.** Use undirected edges for physical or co-expression links, and directed edges where one gene regulates another.
5. **Store the network** as an edge list or adjacency matrix, then import it into Cytoscape or STRING.
6. **Map attributes visually.** For example, node color shows up- or down-regulation, node size shows degree, and edge thickness shows confidence.

Example: ERBB2 is linked to GRB2 and PIK3CA, and PIK3CA is linked to AKT1. This gives the nodes ERBB2, GRB2, PIK3CA and AKT1, with three edges between them.

Conclusion: Representing genes as nodes and interactions as edges allows the use of topological measures (degree, betweenness, clustering) to find key genes and functional modules in breast cancer.

Identifying Hub Proteins Using Degree Centrality (4 Marks)

Concept: Degree centrality is the number of direct connections a node has. A **hub protein** is a node with a degree much higher than the network average. Hubs are often essential proteins, and their removal can disrupt the network.

Formula:

- Degree of node *i*: k(i) = number of edges attached to *i*.
- Normalized degree centrality: C_D(i) = k(i) / (N − 1), where N is the total number of nodes.
- For directed networks, the **in-degree** (incoming links) and **out-degree** (outgoing links) are calculated separately.

Steps:

1. **Build the PPI network** by importing the dataset into Cytoscape (nodes = proteins, edges = interactions).
2. **Filter weak interactions** (e.g., confidence score below 0.7).
3. **Calculate degree** for every node, using *Tools → NetworkAnalyzer → Analyze Network*.
4. **Rank proteins** from the highest to the lowest degree.
5. **Set a threshold**, such as the top 5–10% of nodes or those with degree above the mean plus one standard deviation.
6. **Confirm the hubs** with the degree distribution plot, which typically follows a power law (few hubs, many low-degree nodes). Cross-check with cytoHubba.
7. **Validate biologically** with enrichment analysis, literature, or survival data.

Example:

Protein	Degree	Interpretation
TP53	45	Hub
AKT1	38	Hub
GRB2	12	Moderate
Peripheral protein	2	Non-hub

Conclusion: Proteins with the highest degree centrality are identified as hubs. They are likely to be functionally important and are good candidates for drug targets or biomarkers. Degree alone ignores bottlenecks, so it is often combined with betweenness centrality.

Representing Directionality from a KEGG Pathway (5 Marks)

Concept: KEGG pathways show signal flow, such as activation, inhibition and phosphorylation, with arrows. An undirected graph loses this information, so a **directed graph (digraph)** is used.

Representation:

- Each edge becomes an **ordered pair (A → B)**, meaning A acts on B. The edge has a source node and a target node, drawn with an arrowhead.
- Edge direction follows the order of signal transfer. For example, EGFR → GRB2 → SOS1 → KRAS → RAF1 → MEK → ERK.

- **Edge attributes** store the interaction type: activation (+), inhibition (−), phosphorylation, or binding.
- **Visual mapping:** an arrow for activation, a blunt T-shaped end for inhibition, and a dashed line for indirect effects.

Steps:

1. **Obtain the pathway** from KEGG (e.g., hsa04010, MAPK pathway) as a KGML file.
2. **Import into Cytoscape** using KEGGscape or CyKEGGParser. The relation tags (activation, inhibition) become edge attributes.
3. **Define the edge list** with three columns: *Source, Target, Interaction type*.
4. **Store as an adjacency matrix.** For a directed graph the matrix is **not symmetric**: A[i][j] = 1 means i → j, but A[j][i] may be 0. Signs (+1/−1) can show activation or inhibition.
5. **Apply a directed layout** (hierarchical layout) so the signal reads from receptor to transcription factor.
6. **Analyze with directed measures:** in-degree, out-degree, directed paths, and feedback loops (cycles).

Interpretation: In-degree identifies receiving nodes (e.g., transcription factors). Out-degree identifies signaling sources (e.g., kinases or receptors). Directed paths show how a signal travels from a receptor to its target.

Conclusion: A directed, attribute-rich graph keeps the cause-and-effect logic of a KEGG pathway, which an undirected graph cannot do. This enables signal-flow and feedback analysis.

Representing Reversible Reactions in a Metabolic Network (5 Marks)

Concept: A metabolic reaction converts substrates into products. Some reactions are **irreversible** (one direction only), while others are **reversible** (they can proceed in both directions depending on conditions). The network model must show this difference.

Representation options:

Model	How reversibility is shown
Directed graph	An irreversible reaction is one arrow (A → B). A reversible reaction is **two opposite arrows** (A ⇄ B), i.e., edges A → B and B → A.
Undirected edge	A single line between A and B is a simple option when direction is not needed.
Bipartite graph	Two node types (metabolites and reactions). Metabolite → reaction → metabolite. A reversible reaction has bidirectional edges, or is split into forward and reverse reaction nodes.
Stoichiometric matrix (S)	Rows are metabolites and columns are reactions. Substrates are negative and products are positive. A reversible reaction has a flux that can be positive or negative ($-\infty < v < +\infty$).

Steps:

1. **Collect reactions** from KEGG, BioCyc or a genome-scale model (e.g., Recon3D).

2. **Mark each reaction** as reversible or irreversible using the database annotation or Gibbs free energy (ΔG).
3. **Create nodes** for metabolites (and reaction nodes in a bipartite model).
4. **Add edges:** one directed edge for irreversible reactions and two opposite edges for reversible ones.
5. **Attach attributes:** enzyme (EC number), reversibility flag, flux bounds, and cofactors.
6. **Import into Cytoscape** and use a visual style to distinguish reversible edges, for example with double arrowheads or a different color.

Example: The glycolysis step glucose-6-phosphate ⇄ fructose-6-phosphate (catalyzed by phosphoglucose isomerase) is reversible and gets two arrows. The step catalyzed by hexokinase (glucose → glucose-6-phosphate) is irreversible and gets one arrow.

Conclusion: Reversible reactions are best modeled as bidirectional edges (or flux with both signs). This preserves metabolic flexibility and allows the use of flux balance analysis and pathway analysis.
